

HEALTHCARE ANALYTICS FOR DOCTOR VISITS

Complete Data Analytics & Machine Learning Project Report

Dataset: Healthcare Doctor Visits | 5,190 records | 13 columns

1. Project Objective

Analyze patient-related information and identify useful patterns associated with the number of doctor visits. The main question is: Why do some people visit a doctor more often than others?

2. Dataset Overview

The dataset contains 5,190 records and 13 columns. The target variable is 'visits'. 'Unnamed: 0' is treated as a record identifier.

3. Data Preparation

The CSV was loaded and validated. Missing values, data types and unique values were checked. No missing values were found. The identifier column was excluded from predictive modeling. An 80/20 train-test split with random seed 42 was used.

4. Exploratory Data Analysis

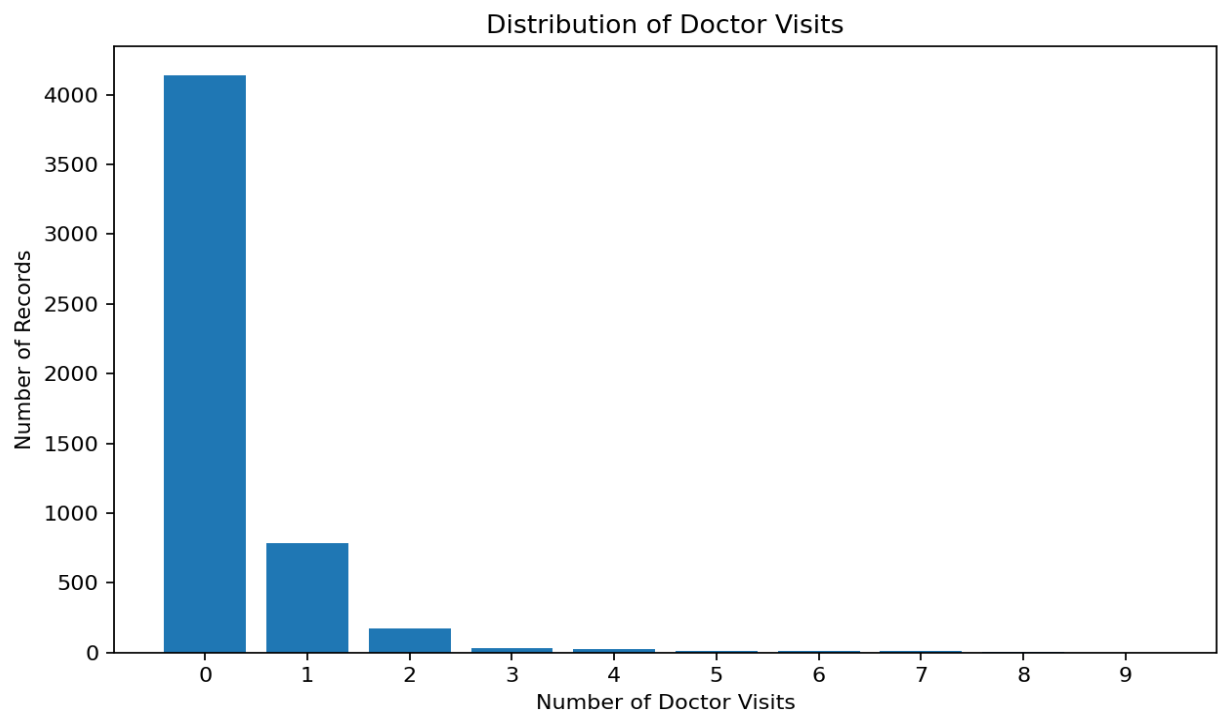


Figure 1. Distribution of Doctor Visits

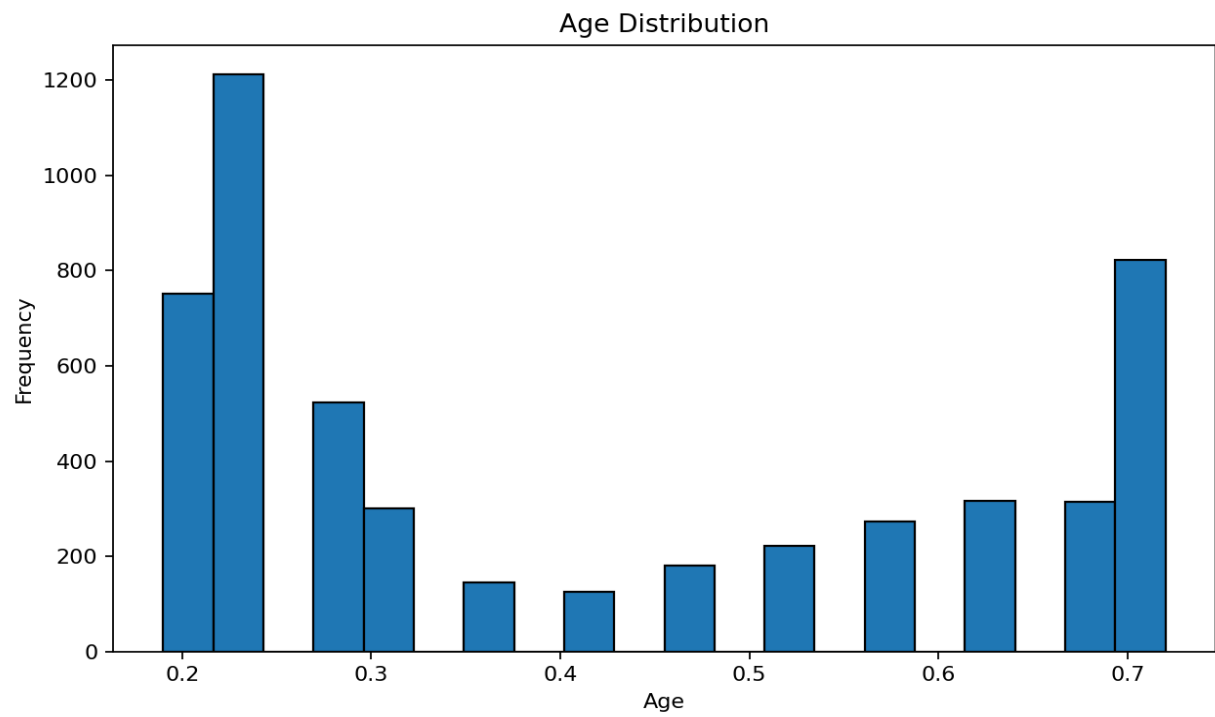


Figure 2. Age Distribution

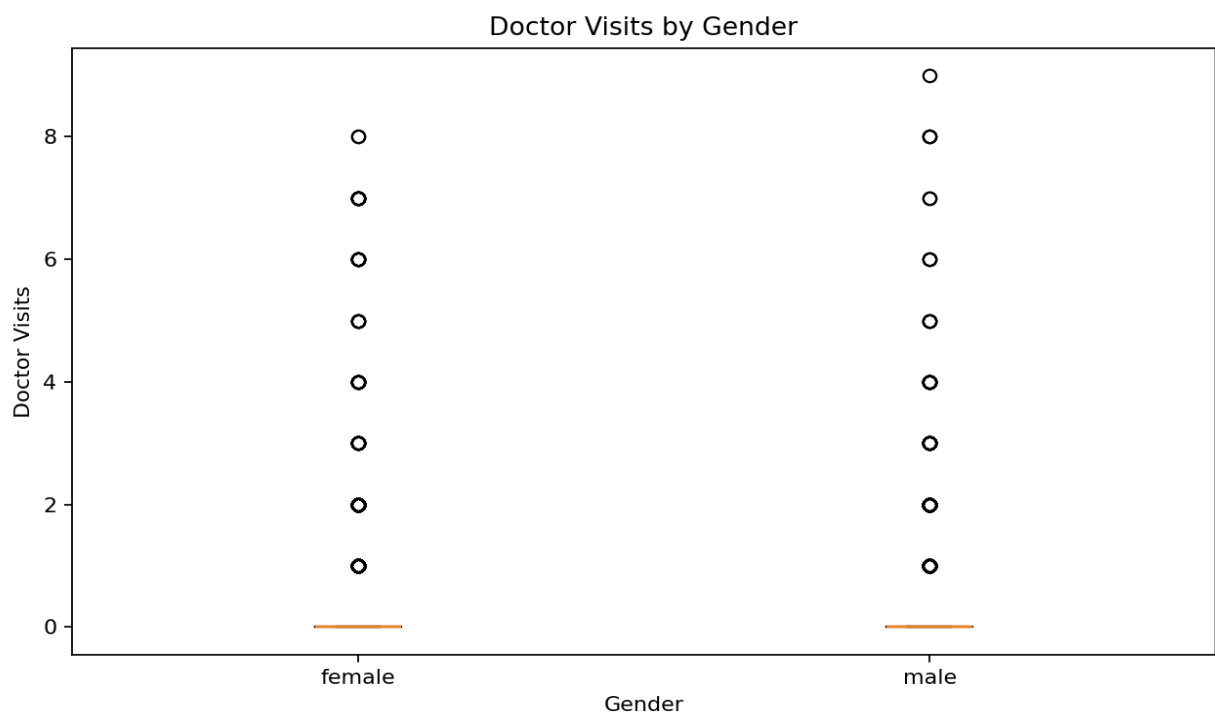


Figure 3. Doctor Visits by Gender

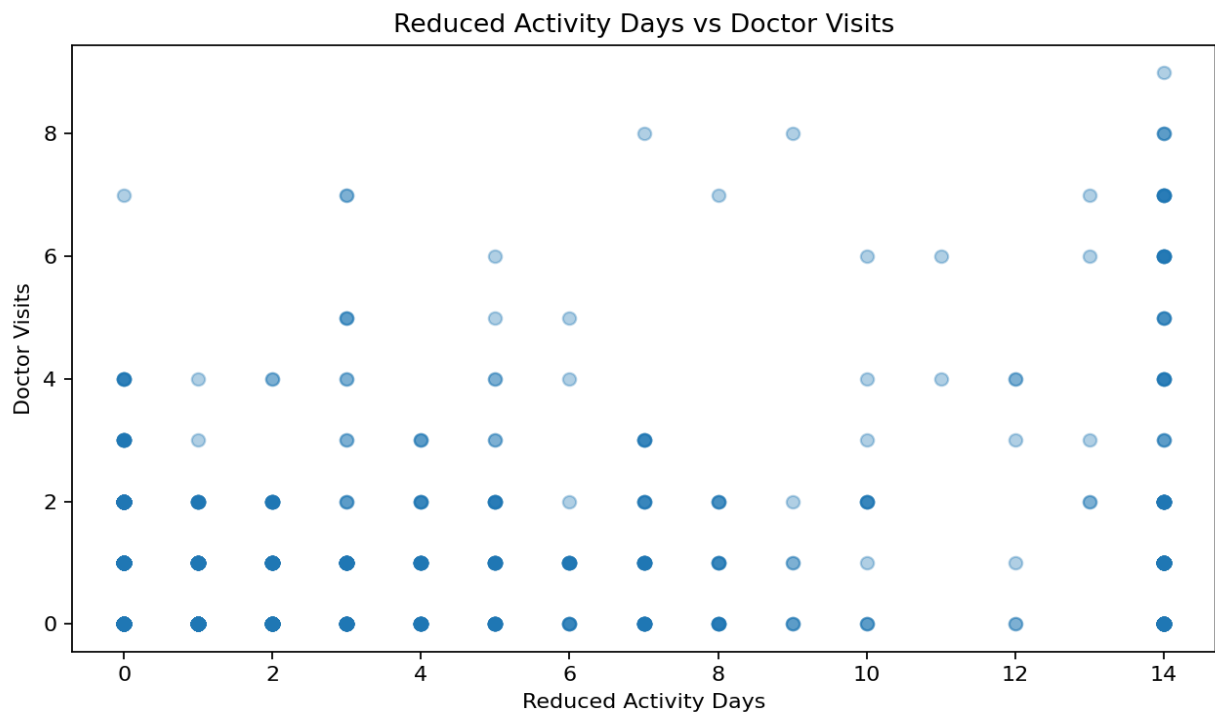


Figure 4. Reduced Activity Days vs Doctor Visits

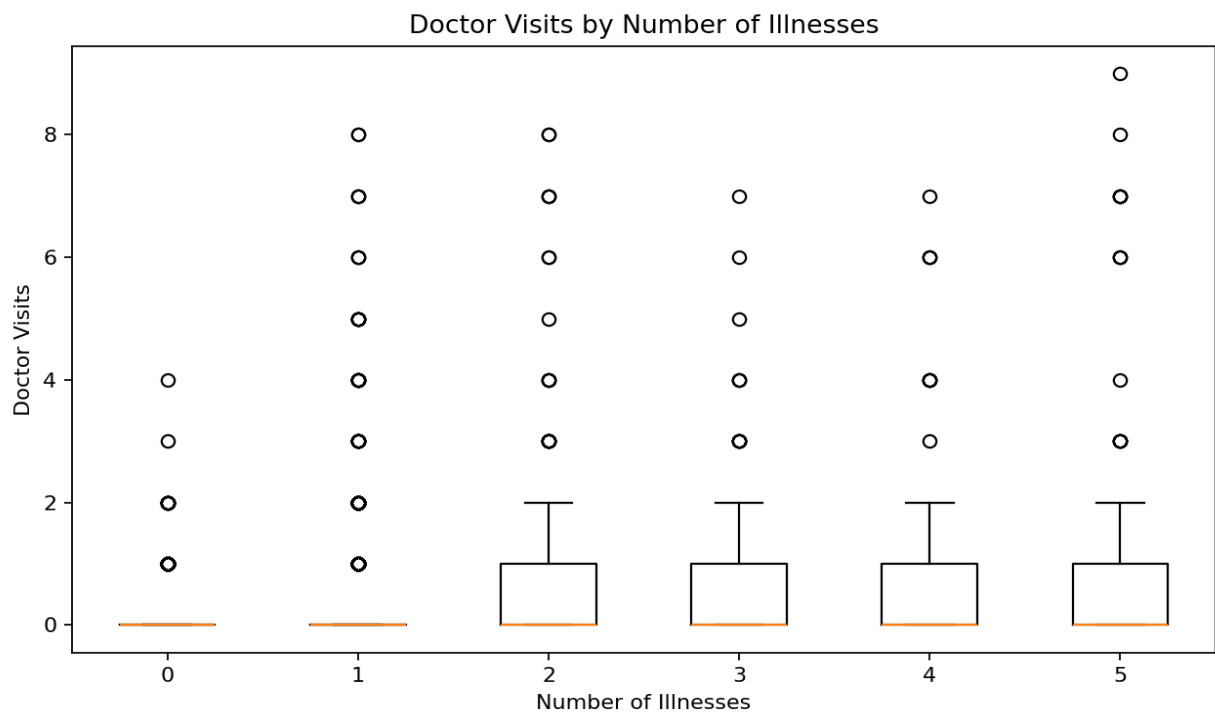


Figure 5. Doctor Visits by Number of Illnesses

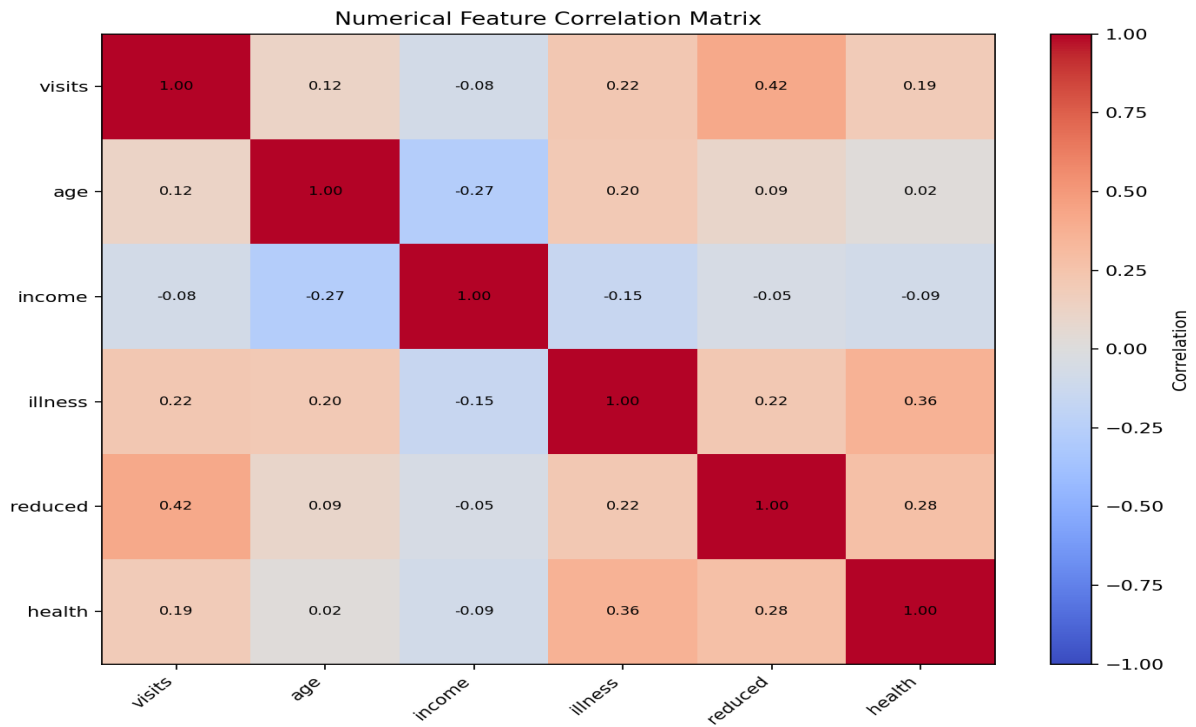


Figure 6. Numerical Feature Correlation Matrix

5. Key Observations

The average number of doctor visits is approximately 0.30. Visits are right-skewed with many zero-visit records. Reduced activity days show the strongest numerical association with visits in this dataset, followed by illness and health-related variables.

6. Machine Learning Results

Model	MAE	RMSE	R ²
Ridge Regression	0.4202	0.8348	0.2118
Random Forest	0.4193	0.8589	0.1656

7. Dashboard and Conclusion

The project includes a Streamlit dashboard with filters, KPIs, charts, healthcare-support summaries and filtered-data download. The complete workflow covers data validation, EDA, statistical association analysis, machine learning, evaluation and visualization.

Note: These are statistical associations in the supplied dataset, not causal or medical conclusions.